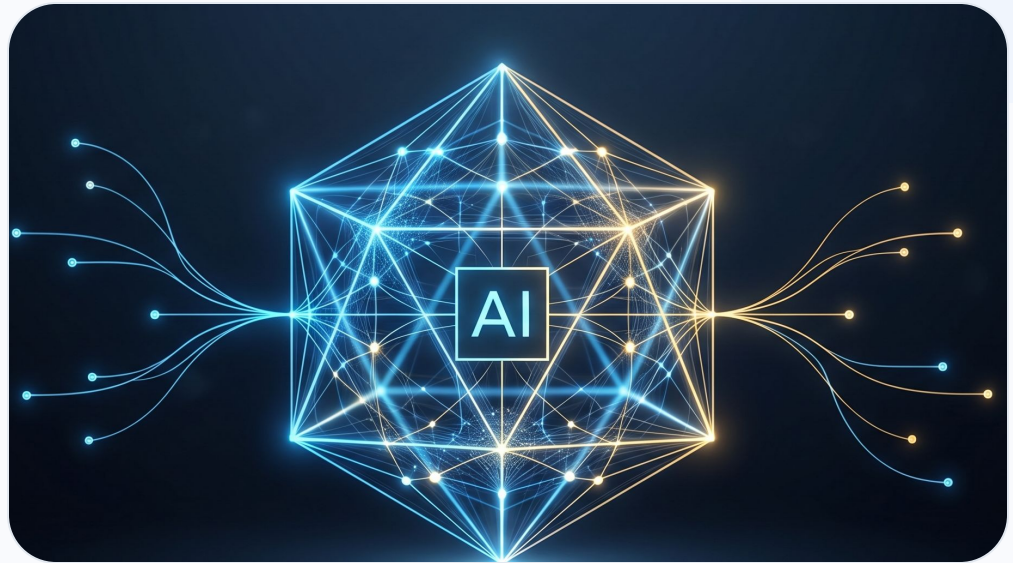


Demystifying AI: Plain & Simple

Ann Catherine Jose

13 June 2026

San Jose Public Library
Willow Glen Branch



Who am I

Software Engineer. Hands-On Tech Enthusiast.

Former **Director of Engineering** at **Intuit**, leading QuickBooks and Gen AI

26 years building reliable software and leading engineering teams at Intuit, Philips, SAP

Currently building useful software applications and teaching AI to the community.

annjose.com | latentsignal.org



Ann Catherine Jose

Agenda

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Intro to AI

- › AI for you
- › What is AI
- › How it works

02

AI in action

- › Demo
- › How I use AI
- › Good tools

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Risks of AI

- › AI gets wrong
- › Use AI safely
- › Smart use

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Wrap up

- › Future of AI
- › 3 takeaways
- › Q & A

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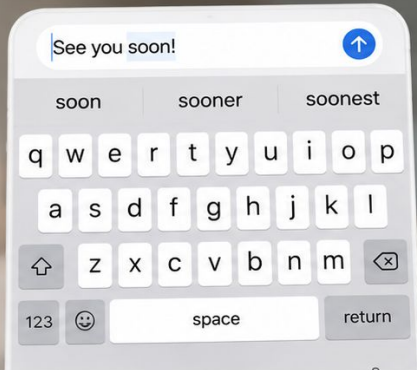
What does AI mean to you?



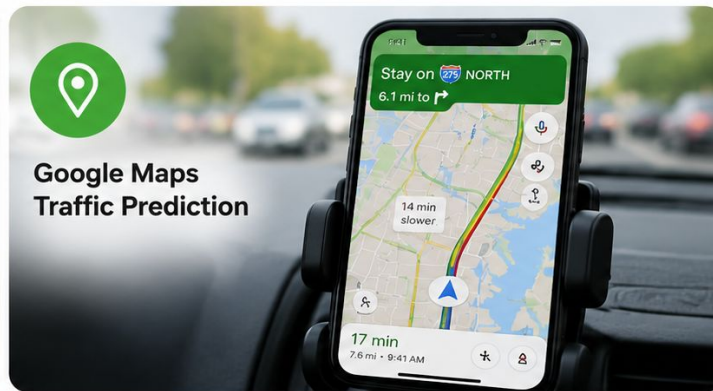
You've used AI before ChatGPT



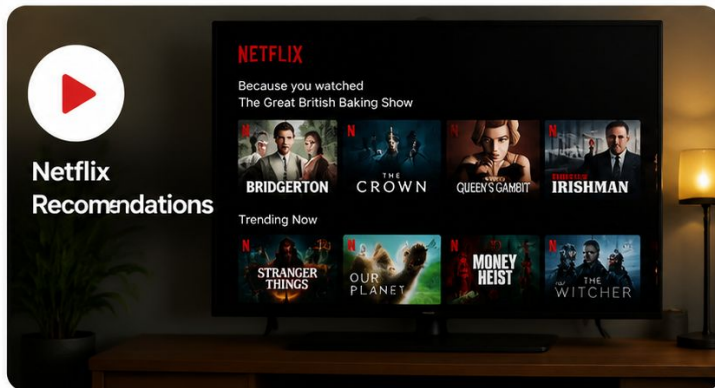
Autocorrect & Predictive Text



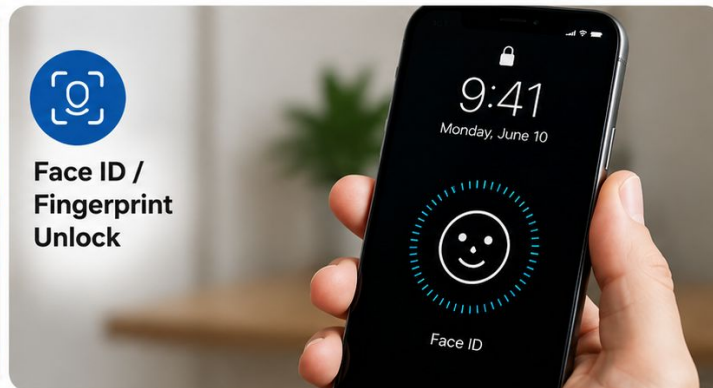
Google Maps Traffic Prediction



Netflix Recommendations



**Face ID /
Fingerprint
Unlock**



5 things AI can help you with today

Understanding a confusing letter

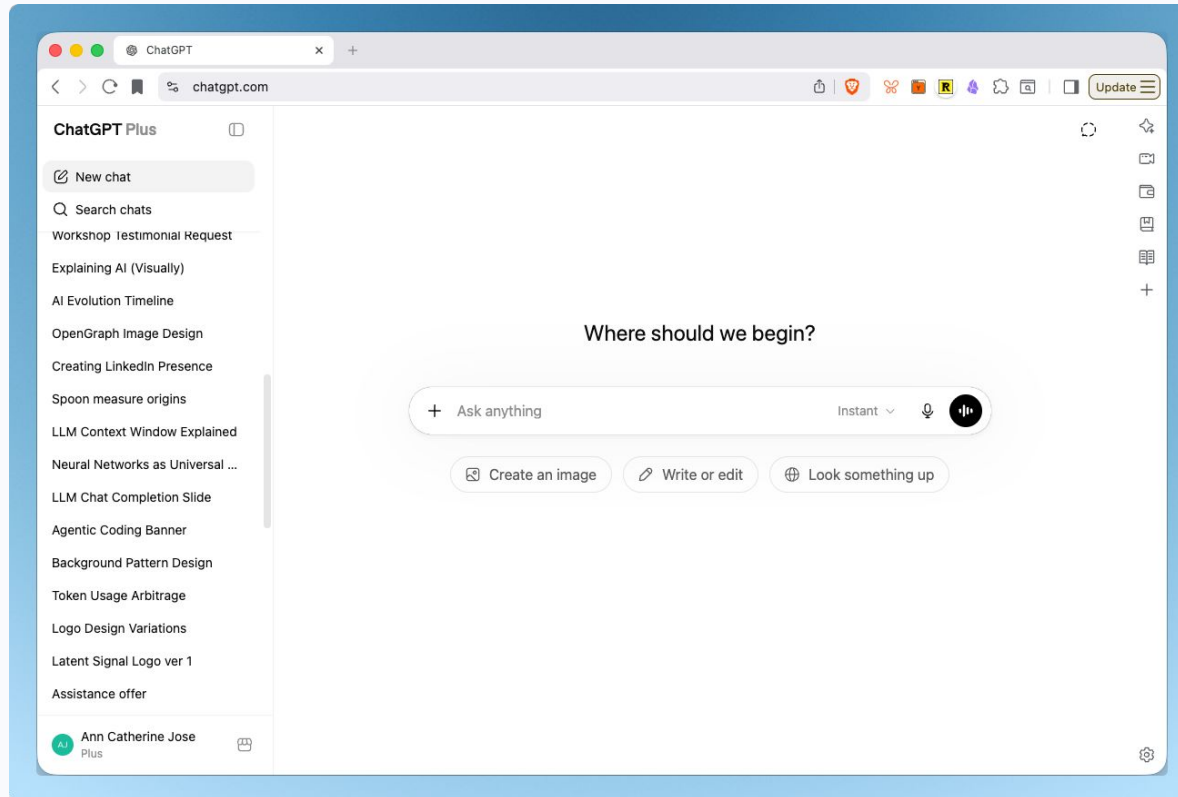
Plan a trip

Learn about a new topic

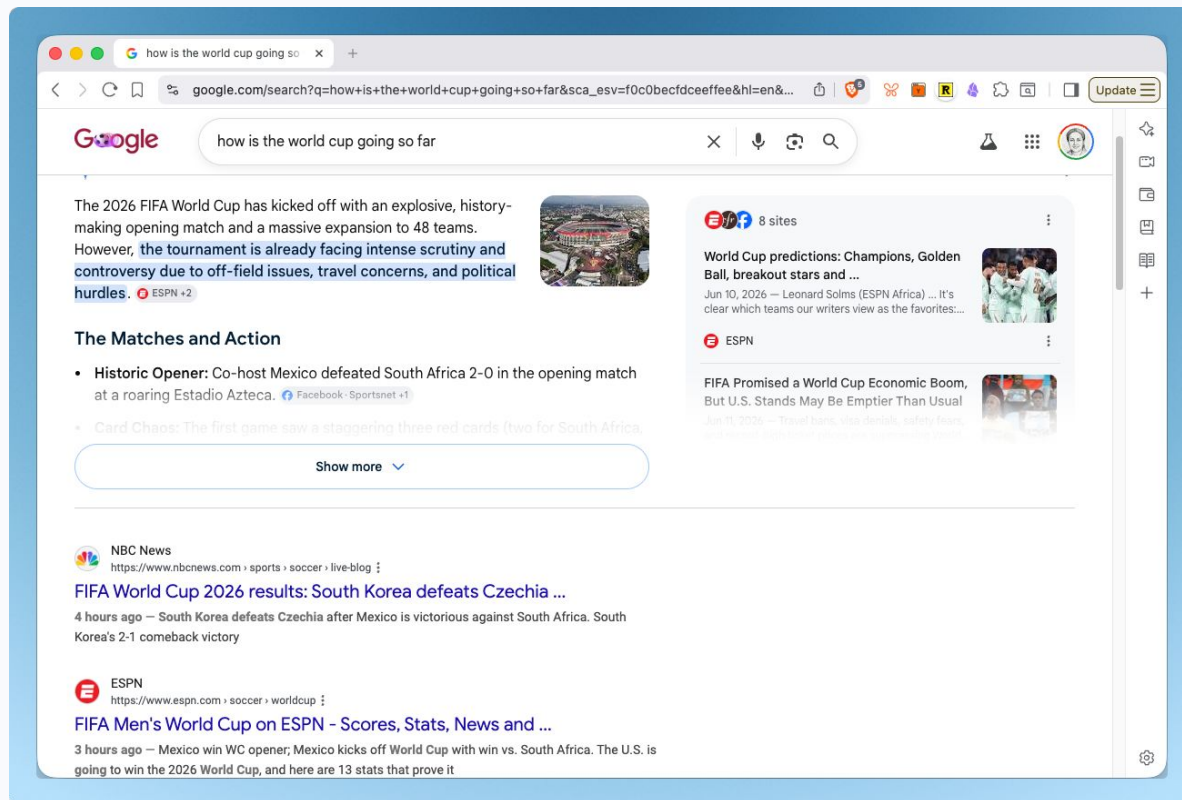
Troubleshoot a phone / computer problem

Understand a complex process

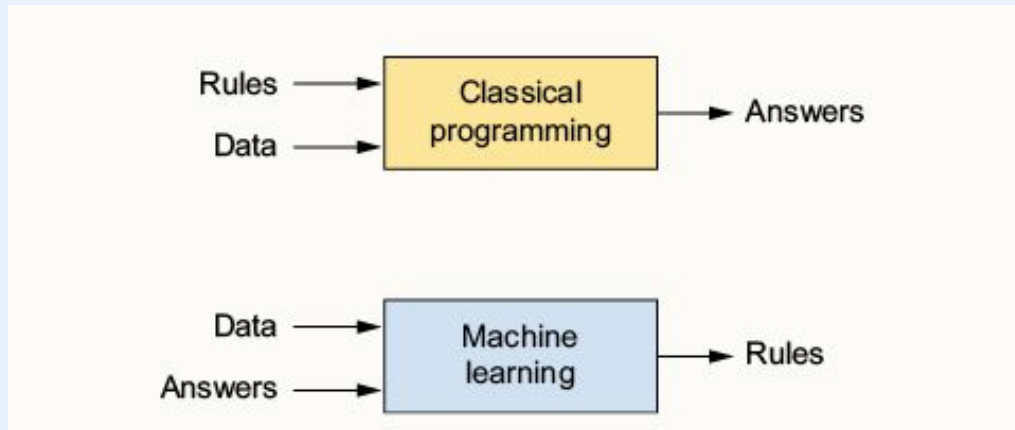
Demo - ChatGPT



Demo - Google AI Overview



What is AI?



AI is computer program that **learns patterns** from **massive** amounts of **data**, and make **predictions** about data it **hasn't seen before**.

How AI learns

A human child

The parents point at a dog and say ‘this is a doggie’, repeat this many times, different variations.

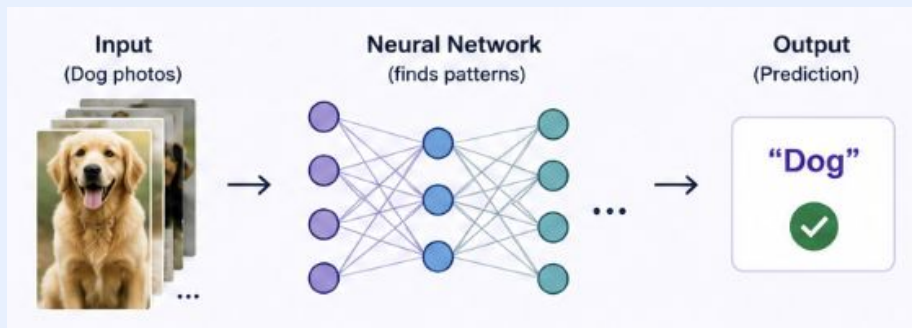
Eventually, the child gets it and recognizes a dog

A machine

Show a machine 100,000 pictures of a dog, each one tagged as ‘this is a dog’. It finds the underlying pattern.

Eventually, the machine ‘learns’ the general pattern

Neural Networks: The technology behind modern AI



ChatGPT: Same tech, different data

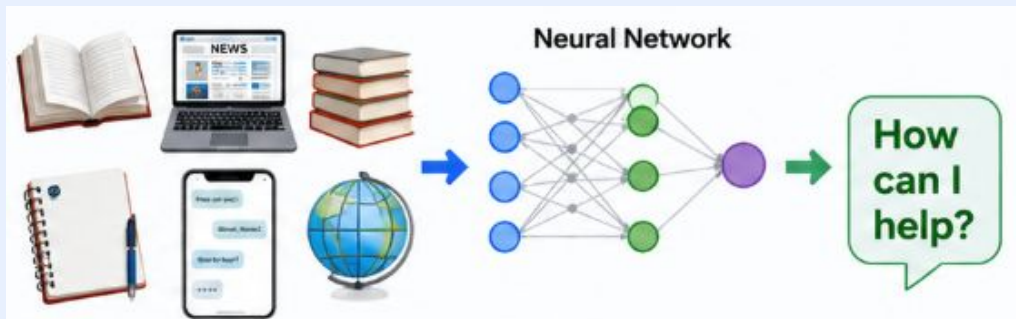
Earlier AI

- ❖ Learns specific patterns from pictures
- ❖ Recognizes objects

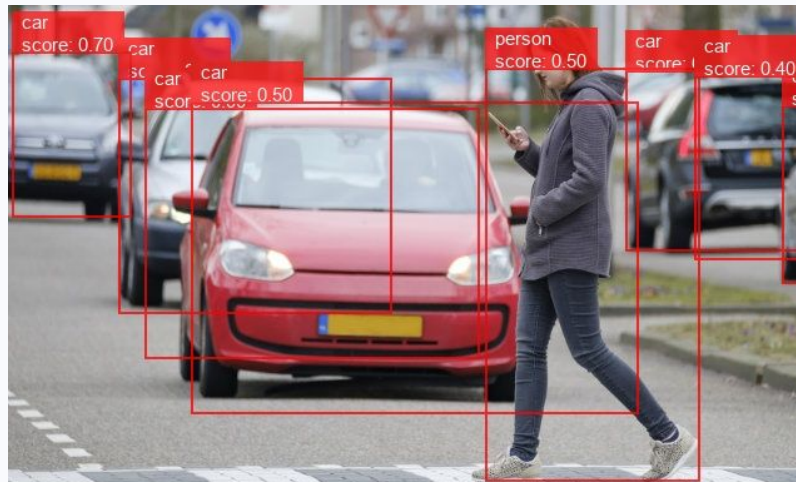
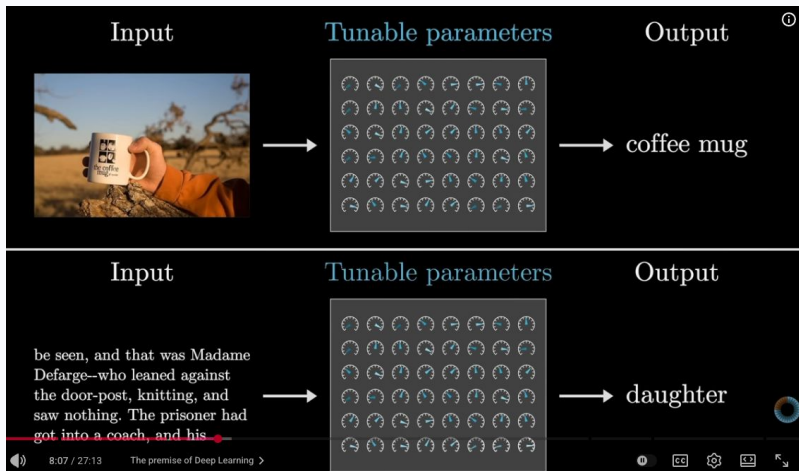
Generative AI (ChatGPT)

- ❖ Learns generic patterns from all digital content
- ❖ Generates text

ChatGPT learns patterns in language and is fine-tuned to sound helpful.



Practical Applications



Neural networks can learn patterns in images, speech, text and more.

Large Language Models (LLMs)



Foundation Model

A model is trained on vast amounts of text, images, audio, video (all digital content).

Its main job: predict the next word

PROMPT:

"The sky is _____"



Chat bot (Assistant)

Fine-tuned to answer questions, be helpful, confident. One word at a time:

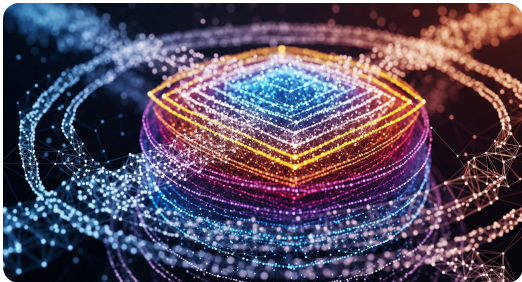
"The sky is blue"

"The sky is blue because _____"

"The sky is blue because sunlight _____"

LLM is a next token predictor. [GPT-2 Demo - Transformer Explainer](#) →

AI Breakthroughs

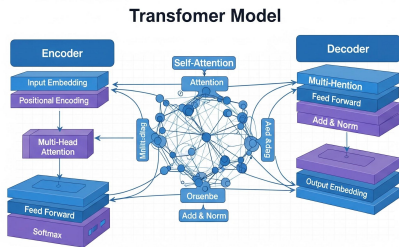


2012

A New Kind of Brain

Deep Learning

Researchers trained complex neural networks that could learn from examples at scale.



2017

A New Kind of Engine

The Transformer (The "T" in GPT)

Google researchers invented an architecture optimized for understanding sequences.



2022

A New Kind of Scale

ChatGPT Launch

OpenAI proved that massive compute and data yield emergent reasoning. They made AI available for everyone.

Building an AI system

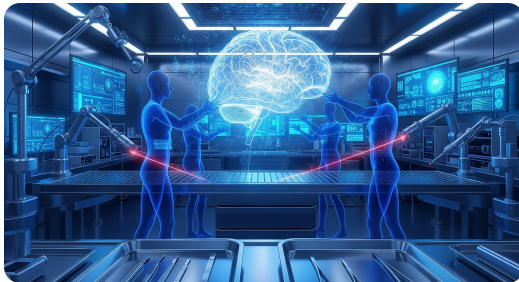


STAGE 01

Training

Knowledge Foundation

Massive ingestion of digital content - everything on the internet: books, articles, websites, and papers to learn patterns.



STAGE 02

Post-training

Alignment & Safety

Fine-tuning for specific tasks and human alignment. This stage implements safety guardrails ensures the model follows instructions.



STAGE 03

Inference

Live Interaction

The operational phase where you ask questions in ChatGPT and the model generates responses in real-time. Speed and cost are important.

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Popular AI Tools



Chat GPT

Open AI

ask questions
help write prose
general purpose chat
create images
write code

chatgpt.com

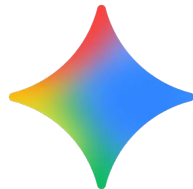


Claude

Anthropic

ask questions
general purpose chat
create images
write code

claude.ai



Gemini

Google

image generation,
integrated into
Google products,
Presentations, Docs

gemini.google



Perplexity

Perplexity

knowledge gathering
search with citations
real-time info
computer use

perplexity.ai



Notebook LM

Google

understand anything
learn topics
read books in depth
mind maps
quiz

notebooklm.google

Claude – Compare travel options

THE PROMPT

I am retired. Compare a cruise vacation, a guided bus tour, and an independent trip to Alaska.

Compare:

- cost
- physical effort
- convenience
- flexibility
- overall experience

Put the answer in a table.

Claude is available at: <https://claude.ai>

Perplexity - Local News

THE PROMPT

What are the 5 most important news stories in San Jose today?

For each story:

- Give a one-sentence summary
- Explain why it matters
- Cite your source

Perplexity is available at: <https://perplexity.ai>

Gemini - Explain the document

THE PROMPT

(upload a PDF)

"Review this notice and answer these questions:

- What is this notice about?
- What does it mean for me?
- Do I need to do anything?
- Is there a deadline/fee/rate change?

CONSTRAINTS (RULES)

- Use only the uploaded notice
- Do not guess or add outside info
- Fallback: "Not stated"
- Include quote + page number for each point

Gemini is available at: <https://gemini.google.com>

Create an image - Gemini

THE PROMPT

Create a realistic image of Willow Glen Library in the year 2050.

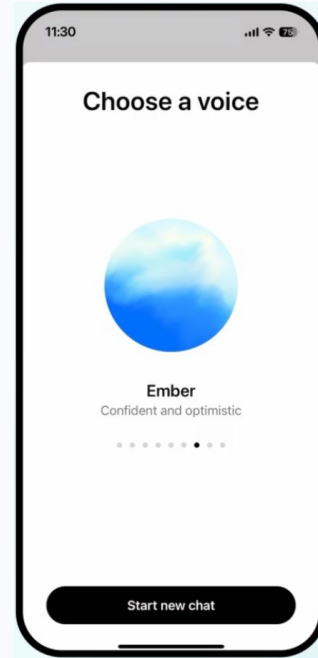
Keep it recognizable but show how transportation and technology might evolve.

Gemini is available at: <https://gemini.google.com>

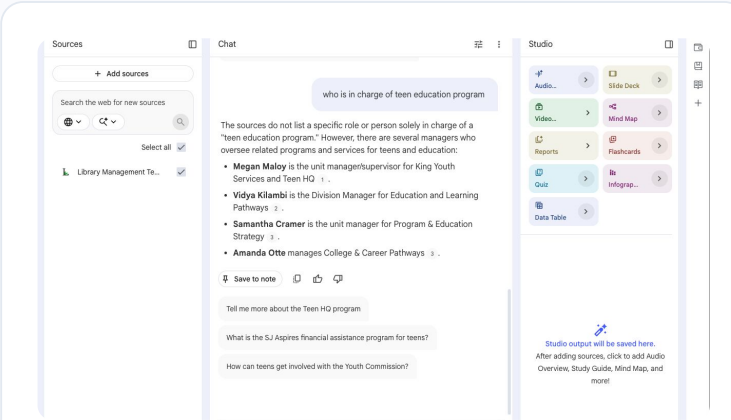
Talk to AI - ChatGPT Voice Mode

How to use Voice Mode

- Install ChatGPT app on your phone (iOS / Android)
- Open ChatGPT app
- Click voice button next to the mic button.
- (The button looks like a sound wave.)
- ChatGPT will be in voice mode
- Now you can talk to ChatGPT

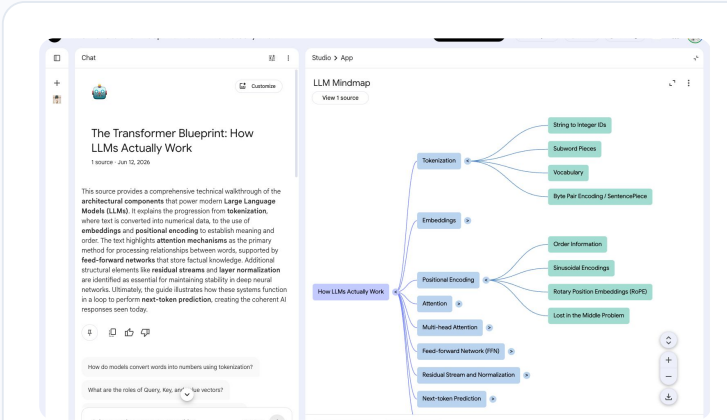


Learn with Notebook LM



Information Retrieval

Upload your documents to create a personalized AI research assistant that knows your specific sources.



Visual Mind Mapping

Synthesize complex information into structured visual maps to better understand connections and themes.

Notebook LM is free and available at: <https://notebooklm.google.com>

How I use AI

01 Plan a day trip to Napa for my birthday

02 Compare flight tickets to India for next month

03 How to do something on my computer or phone

04 Brainstorm ideas and arguments on a topic

05 Learn concepts - index funds, history

06 Compare products and see pro/cons of each

07 Designing my website

08 Agentic coding - a big portion of my usage

09 Understand an article in depth

10 Preparing for this talk - compare options, review the deck

Create Images



Willow Glen Library: Past & Future



1970: The Community Hub

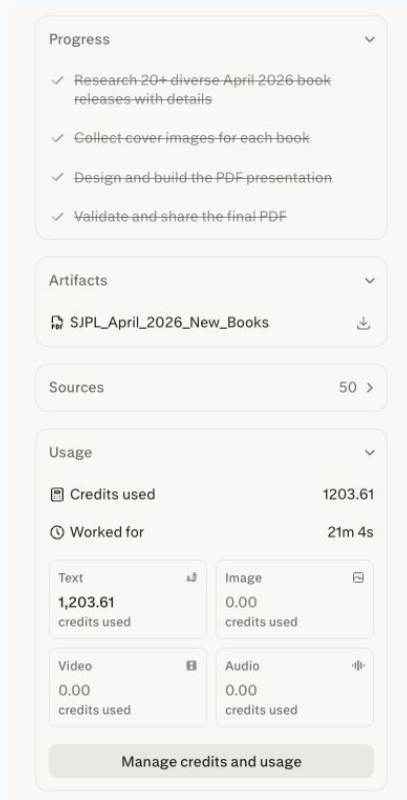
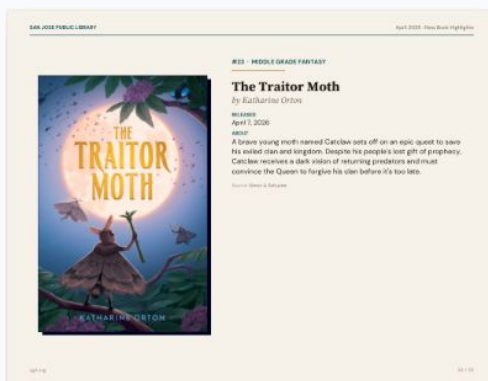
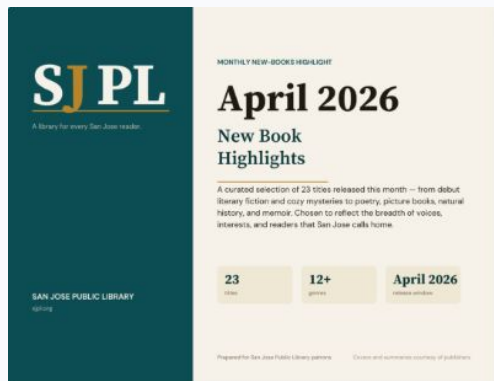
AI PROMPT: A sunny Saturday afternoon in San Jose. Create a highly realistic historical photograph showing families arrive in classic 1970s cars to a bustling branch filled with physical books, newspaper racks, and hand-posted community boards.



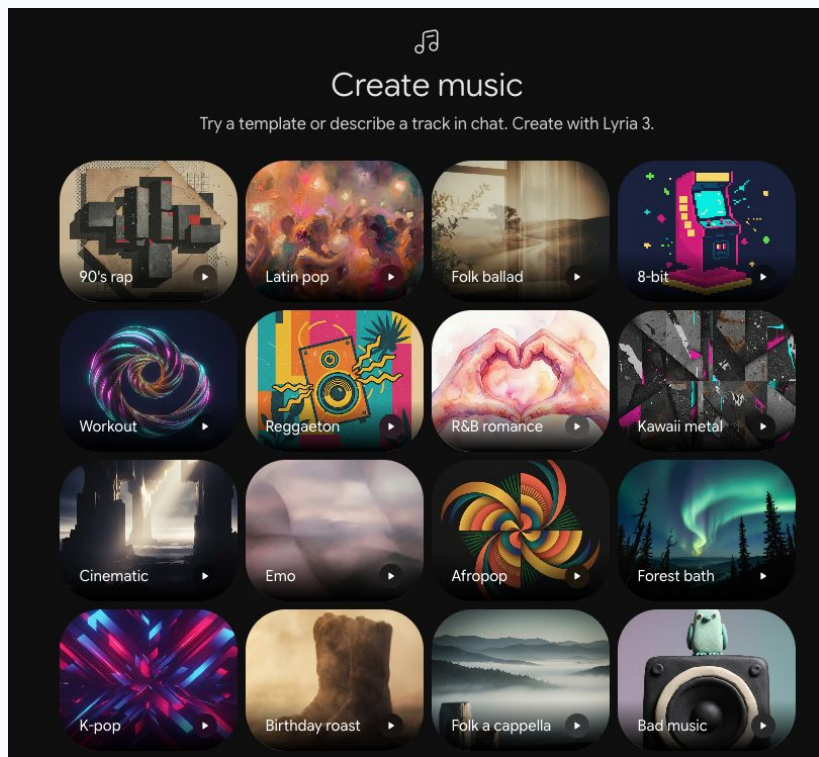
2050: A Vision of Evolution

AI PROMPT: Keep it recognizable as Willow Glen Library, but imagine how technology, autonomous transportation, and digital services might evolve over the next 25 years.

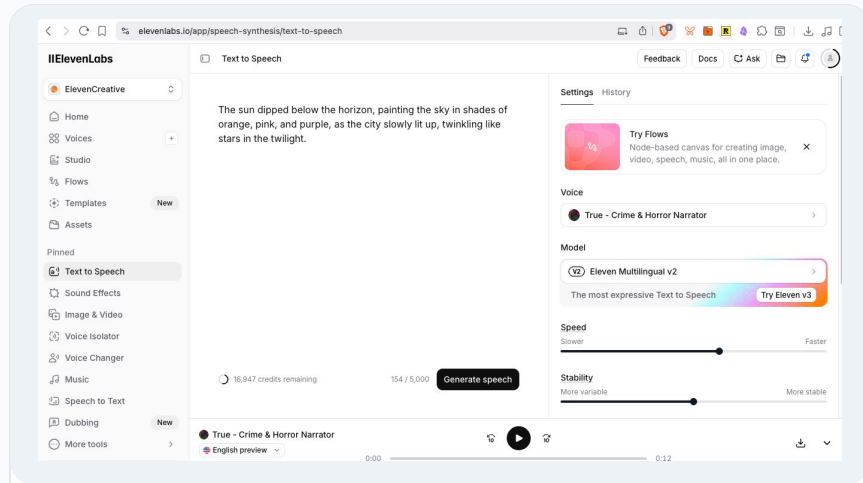
Long running agents - Perplexity



Create Music – Gemini

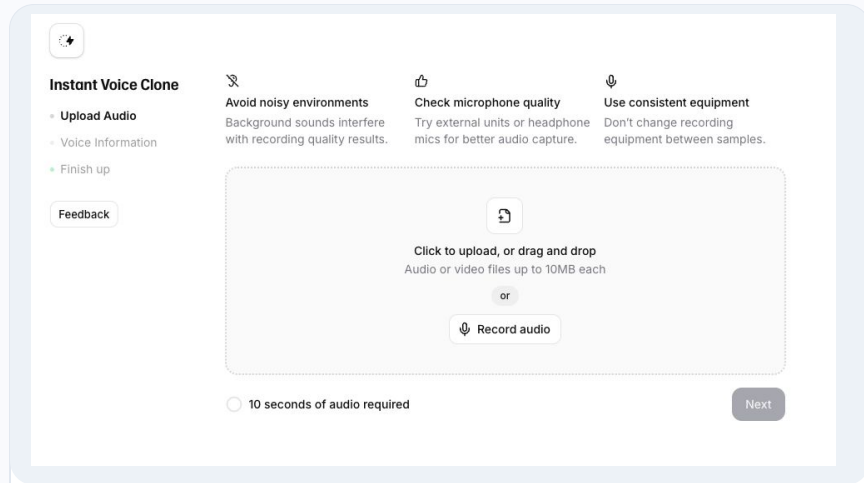


Voice AI



Voice Narration

Provide a detailed description of the intended visual or evidence for this section.



Clone your voice

Use this area to define the prompt or context for the second supporting image.

Putting it altogether



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Risks of AI

The Reality

AI is designed to be **helpful**, not necessarily truthful.

It can sound extremely confident even when it is **wrong**.

It may echo your own biases and agree with you even if you are mistaken.

Do not rely on AI for:



Medical decisions



Financial decisions

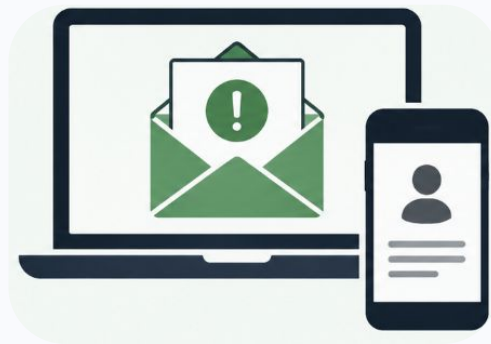


Legal decisions



Emotional support

AI Can be Used to Trick You



01

Impersonation

- ❖ Voice clones
- ❖ Fake grandchild calls
- ❖ Fake customer support

02

Fake content

- ❖ Fake images
- ❖ Deep fake videos
- ❖ Celebrity endorsement

03

Social engineering

- ❖ Phishing email
- ❖ Fake urgent texts
- ❖ Fake password reset

AI makes deception faster, cheaper, and easier – and harder to spot.

How to protect yourself



STOP if:

- Someone asks for money urgently
- Someone asks for gift cards
- Someone asks for cryptocurrency
- Someone asks for your passwords
- Someone says "don't tell anyone"



VERIFY

- Call back using a known number
- Contact family members
- Check with a trusted friend
- Go directly to the official website



SLOW DOWN

- If money is involved, wait 24 hours before acting.

Learn more – [AI Fraud Prevention Guide](#) by Microsoft

Safe use of AI



Protect your privacy

Don't share passwords, SSN, financial account information, or sensitive personal details with AI tools.



Verify before you trust

AI can sound confident and still be wrong. Double-check medical, legal, financial, health advice before acting on it.



Be Careful What You Upload

Anything you upload may be processed by the AI service. Avoid sharing private or confidential documents.



You're Still Responsible

AI can help draft emails, summarize docs, and answer questions, but we are responsible for checking final result.

Smart use of AI

Find Information

- What happened in the news today?
- Compare two products
- Plan a trip to Japan
- Explain a concept
- Verify important facts

Analyze Information

- Summarize this article
- Explain this chart
- Compare these documents
- Describe this image
- Usually more reliable

Better Prompts = Task + Context + Constraints = Better Answers

Critical Thinking

When you read on the internet...

- *Can I verify this from a primary source?*
- *What is the weakest part of this argument?*
- *If this is wrong, how bad is it?*
- *What would a skeptic of this view say?*

When you write using AI...

- *Does this actually sound like me?*
- *Is everything factually correct?*
- *Did it make up any specifics?*
- *Am I okay being responsible for this?*

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Future of AI

Moving Faster Than Ever

AI is improving at an incredible speed. With more money and better technology, we are building AI agents that can complete complex tasks on its own.

Jobs and the Economy

Knowledge work is changing the most with real productivity gains. New types of jobs will appear, but it will be a big adjustment.

Trust and Power

Fake photos and videos make it harder to know what is real. We need to be cautious and enact fair rules to protect our privacy and data.

Cost is increasing

Organizations are seeing the AI bill come due. Free open source models and tools are emerging.

Three Takeaways

1. AI is a tool

Like the internet, a calculator, or a smartphone, AI can be incredibly useful - but it can also cause problems when used carelessly.

Trust, but verify.

2. Don't fall for the hype or the fear

AI is neither magic nor doom.

It's a powerful technology with real strengths, real limitations, and real risks. Stay curious, stay informed of what it can do today.

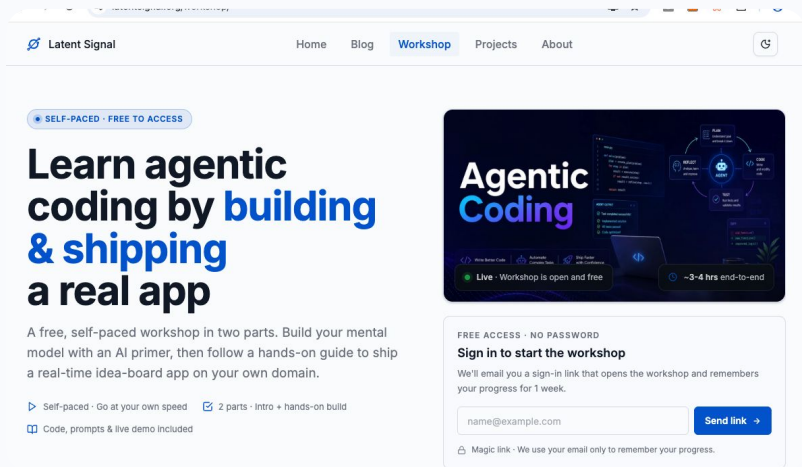
3. Use AI to augment yourself, not replace yourself

The people who benefit most from AI won't be those who avoid it - they'll be those who learn how to use it effectively.

Learn to use AI tools, but focus on what AI can't do: good judgment, human creativity and expertise.

Free AI Workshop

A free hands-on, self-paced online guide to learn how to code with AI, build and ship a real app.



The screenshot shows the homepage of the Latent Signal website. The navigation bar includes links for Home, Blog, Workshop (highlighted), Projects, and About. A badge in the top left corner reads "SELF-PACED · FREE TO ACCESS". The main heading is "Learn agentic coding by building & shipping a real app". Below this, a paragraph describes the workshop as a free, self-paced guide to build a mental model with an AI primer and follow a hands-on guide to ship a real-time idea-board app. A list of features includes "Self-paced - Go at your own speed", "2 parts - Intro + hands-on build", and "Code, prompts & live demo included". On the right, there is a graphic for "Agentic Coding" with a "Live · Workshop is open and free" badge and a "3-4 hrs end-to-end" duration. Below the graphic, a sign-in section titled "FREE ACCESS · NO PASSWORD" asks users to "Sign in to start the workshop" by providing an email address and clicking a "Send link" button. A note mentions a "Magic link" for remembering progress.

Latent Signal Home Blog **Workshop** Projects About

SELF-PACED · FREE TO ACCESS

Learn agentic coding by building & shipping a real app

A free, self-paced workshop in two parts. Build your mental model with an AI primer, then follow a hands-on guide to ship a real-time idea-board app on your own domain.

- Self-paced - Go at your own speed
- 2 parts - Intro + hands-on build
- Code, prompts & live demo included

Agentic Coding

Live · Workshop is open and free

3-4 hrs end-to-end

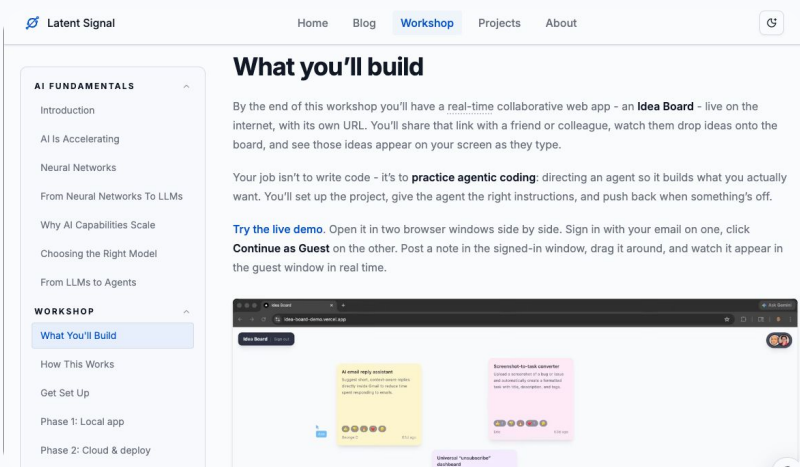
FREE ACCESS · NO PASSWORD

Sign in to start the workshop

We'll email you a sign-in link that opens the workshop and remembers your progress for 1 week.

name@example.com **Send link**

Magic link - We use your email only to remember your progress.



The screenshot shows the "Workshop" page on the Latent Signal website. The navigation bar is the same as the homepage. The page is titled "What you'll build". The main content area is divided into two sections: "AI FUNDAMENTALS" and "WORKSHOP". The "AI FUNDAMENTALS" section lists topics: Introduction, AI Is Accelerating, Neural Networks, From Neural Networks To LLMs, Why AI Capabilities Scale, Choosing the Right Model, and From LLMs to Agents. The "WORKSHOP" section lists topics: What You'll Build (highlighted), How This Works, Get Set Up, Phase 1: Local app, and Phase 2: Cloud & deploy. The "What you'll build" section describes the workshop as a real-time collaborative web app called "Idea Board" and provides instructions on how to try the live demo. Below the text, there is a screenshot of the "Idea Board" interface, which shows a collaborative workspace with sticky notes and a "Universal 'workspace' dashboard".

Latent Signal Home Blog **Workshop** Projects About

What you'll build

By the end of this workshop you'll have a real-time collaborative web app - an **Idea Board** - live on the internet, with its own URL. You'll share that link with a friend or colleague, watch them drop ideas onto the board, and see those ideas appear on your screen as they type.

Your job isn't to write code - it's to **practice agentic coding**: directing an agent so it builds what you actually want. You'll set up the project, give the agent the right instructions, and push back when something's off.

Try the live demo. Open it in two browser windows side by side. Sign in with your email on one, click **Continue as Guest** on the other. Post a note in the signed-in window, drag it around, and watch it appear in the guest window in real time.

AI FUNDAMENTALS

- Introduction
- AI Is Accelerating
- Neural Networks
- From Neural Networks To LLMs
- Why AI Capabilities Scale
- Choosing the Right Model
- From LLMs to Agents

WORKSHOP

- What You'll Build**
- How This Works
- Get Set Up
- Phase 1: Local app
- Phase 2: Cloud & deploy

Idea Board

Universal "workspace" dashboard

Ready to start? Visit the workshop at:
latentsignal.org/workshop

Resources

AI tools we saw today

- **ChatGPT:** chatgpt.com
- **Claude:** claude.ai
- **Gemini:** gemini.google
- **Perplexity:** perplexity.ai
- **Notebook LM:** notebooklm.google
- **Eleven Labs:** elevenlabs.io

Education

Learn about AI

- [Intro videos to neural networks \(3 Blue 1 Brown\)](#)
- [Next token predictor Explainer](#)
- [What is ChatGPT doing \(Stephen Wolfram\)](#)

Security

Stay safe online

- [AARP Fraud Watch Network](#)
- [AI Fraud Prevention](#)
- [Microsoft Cybersecurity Awareness Series](#)
- [FTC Consumer Advice \(scams, fraud\)](#)

Thank you



Ann Catherine Jose

Hands-On Tech Enthusiast.
Software Builder.
Full-Stack Developer.

Questions? Reach out to me...

✉ hello@annjose.com

🌐 annjose.com | latentsignal.org

+👤 linkedin.com/in/annjose